

Anti-parkinsonian agents

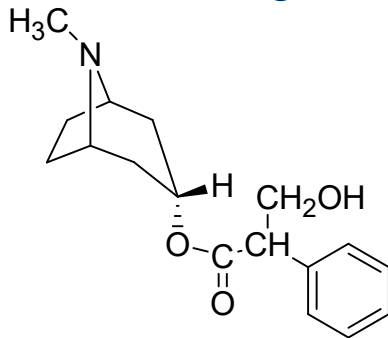


Anti-parkinsonian agents

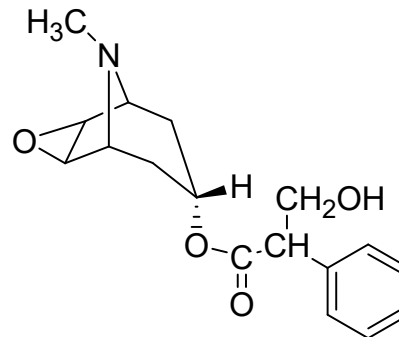
- Parkinson disease is a slowly progressive degenerative neurologic disease
 - Tremor, rigidity, sluggish neuromuscular response and postural instability
- In Parkinsonism dopaminergic input is deficient and cholinergic output remain unchanged
- Antiparkinsonian agents are either **anticholinergic** or **dopaminergic**

Anticholinergics....

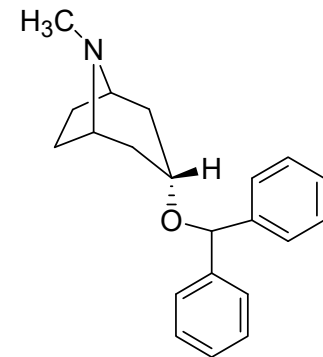
- Anticholinergics used



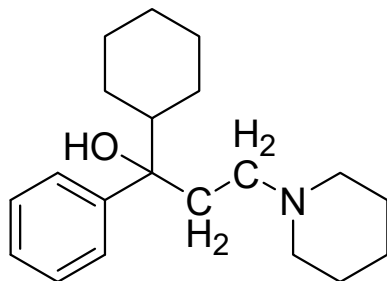
Atropine



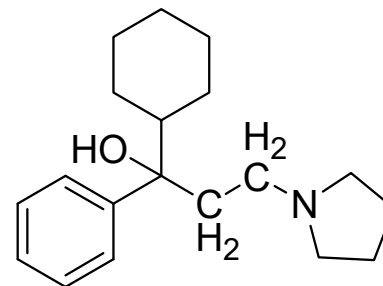
Scopolamine



benzotropine



trihexyphenidyl



procyclidin

Dopaminergic therapy

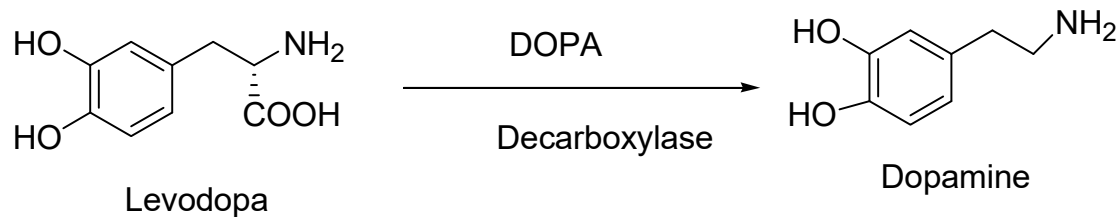
Dopaminergic therapy

- Several approaches are used to eliminate dopamine deficiency in striatum
 - Augmentation of brain synthesis of dopamine
 - Presynaptic dopamine release stimulation
 - Direct stimulation of dopamine receptor

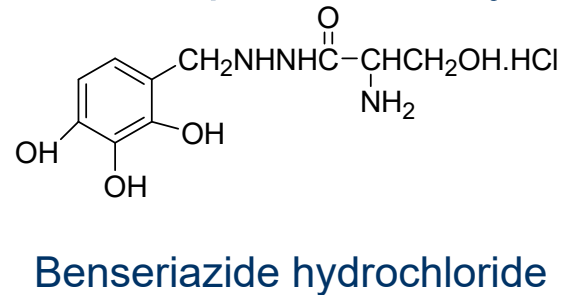
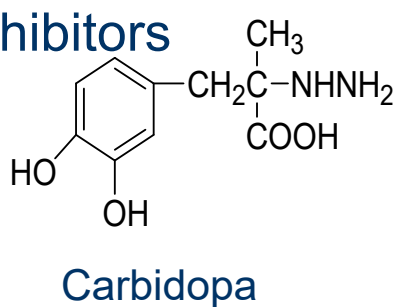
Dopaminergic ...

1. Augmentation of dopamine synthesis

- Levodopa therapy

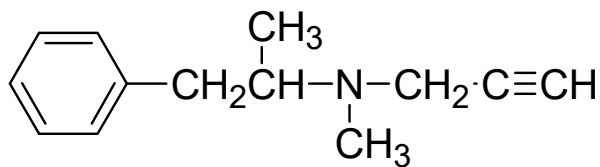


- Levodopa is co-administered with dopadecarboxylase inhibitors



Dopaminergic ...

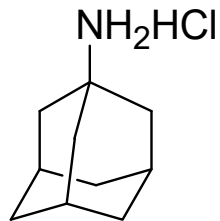
- Another means of increasing dopamine level is inhibition of MAO-B
- Selegiline is selective inhibitor of MAO type B and prevents breakdown of dopamine selectively



Dopaminergic ...

2. Stimulation dopamine release

- Agents that release dopamine from neuronal storage can be used
- Amantidine is an example of such drug

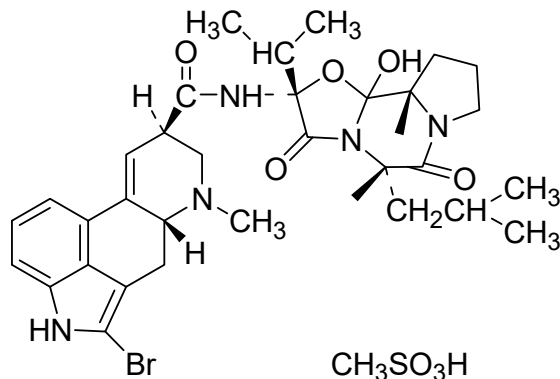


- Amantadine increase dopamine levels at postsynaptic receptor by enhancing release

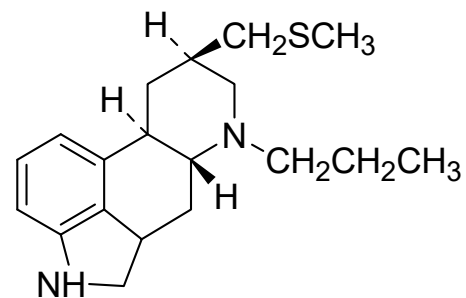
Dopaminergic ...

3. Direct Dopamine agonist

- Includes the ergot alkaloid derivatives **Bromocriptine** and **Pergolide**
- Dopamine agonists have the advantage in that their effect is independent of striato nigral degeneration



Bromocriptine



Pergolide